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Basic concepts

1.1 History of the term “plasma”

In the mid nineteenth century the Czech physiologist Jan Evangelista Purkinje introduced use of the Greek word *plasma* (meaning “formed” or “molded”) to denote the clear fluid that remains after removal of all the corpuscular material in blood. About half a century later, the American scientist Irving Langmuir proposed in 1922 that the electrons, ions, and neutrals in an ionized gas could similarly be considered as corpuscular material entrained in some kind of fluid medium and called this entraining medium plasma. However it turned out that, unlike blood where there really is a fluid medium carrying the corpuscular material, there actually is no “fluid medium” entraining the electrons, ions, and neutrals in an ionized gas. Ever since, plasma scientists have had to explain to friends and acquaintances that they were not studying blood!

1.2 Brief history of plasma physics

In the 1920s and 1930s a few isolated researchers, each motivated by a specific practical problem, began the study of what is now called plasma physics. This work was mainly directed towards understanding (i) the effect of ionospheric plasma on *long-distance short-wave radio* propagation and (ii) *gaseous electron tubes* used for rectification, switching, and voltage regulation in the pre-semiconductor era of electronics. In the 1940s Hannes Alfvén developed a theory of hydromagnetic waves (now called Alfvén waves) and proposed that these waves would be important in astrophysical plasmas. In the early 1950s large-scale plasma physics based *magnetic fusion energy* research started simultaneously in the USA, Britain, and the then Soviet Union. Since this work was an offshoot of thermonuclear weapon research, it was initially classified but, because of scant progress in each country’s effort and the realization that controlled fusion research was unlikely to be of military value, all three countries declassified their efforts in 1958 and

have co-operated since. Many other countries now participate in fusion research as well.

Fusion progress was slow through most of the 1960s, but by the end of that decade the empirically developed Russian *tokamak* configuration began producing plasmas with parameters far better than the lackluster results of the previous two decades. By the 1970s and 1980s many tokamaks with progressively improved performance were constructed and at the end of the twentieth century fusion break-even had nearly been achieved in tokamaks. International agreement was reached in the early twenty-first century to build the International Thermonuclear Experimental Reactor (ITER), a break-even tokamak designed to produce 500 megawatts of fusion output power. Non-tokamak approaches to fusion have also been pursued with varying degrees of success; many involve magnetic confinement schemes related to that used in tokamaks. In contrast to fusion schemes based on magnetic confinement, inertial confinement schemes were also developed in which high-power lasers or similarly intense power sources bombard millimeter-diameter pellets of thermonuclear fuel with ultra-short, extremely powerful pulses of strongly focused directed energy. The intense incident power causes the pellet surface to ablate and, in so doing, act like a rocket exhaust pointing radially outwards from the pellet. The resulting radially inwards force compresses the pellet adiabatically, making it both denser and hotter; with sufficient adiabatic compression, fusion ignition conditions are predicted to be achieved.

Simultaneously with the fusion effort, there has been an equally important and extensive study of space plasmas. Measurements of near-Earth space plasmas, such as the aurora and the ionosphere, have been obtained by ground-based instruments since the late nineteenth century. Space plasma research was greatly stimulated when it became possible to use spacecraft to make routine *in situ* plasma measurements of the Earth's *magnetosphere*, the *solar wind*, and the magnetospheres of other planets. Additional interest has resulted from ground-based and spacecraft measurements of topologically complex, dramatic structures sometimes having explosive dynamics in the *solar corona*. Using radio telescopes, optical telescopes, Very Long Baseline Interferometry, and most recently the Hubble and Spitzer spacecraft, large numbers of *astrophysical jets* shooting out from magnetized objects such as stars, active galactic nuclei, and black holes have been observed. Space plasmas often behave in a manner qualitatively similar to laboratory plasmas, but have a much grander scale.

Since the 1960s an important effort has been directed towards using plasmas for *space propulsion*. Plasma thrusters have been developed ranging from small *ion thrusters* for spacecraft attitude correction to powerful *magnetoplasmadynamic thrusters* that – given an adequate power supply – could be used for interplanetary

missions. Plasma thrusters are now in use on some spacecraft and are under serious consideration for new and more ambitious spacecraft designs.

Starting in the late 1980s a new application of plasma physics appeared – *plasma processing* – a critical aspect of the fabrication of the tiny, complex integrated circuits used in modern electronic devices. This application is now of great economic importance.

In the 1980s investigations began on *non-neutral plasmas*; these mimic the equations of incompressible hydrodynamics and so provide a compelling analog computer for problems in incompressible hydrodynamics. Another application of non-neutral plasmas is as a means to store large quantities of positrons. In the 1990s studies began on *dusty plasmas*. Dust grains immersed in a plasma can become electrically charged and then act as an additional charged particle species. Because dust grains are massive compared to electrons or ions and can be charged to varying amounts, new physical behavior occurs that is sometimes an extension of what happens in a regular plasma and sometimes altogether new. Both non-neutral and dusty plasmas can also form bizarre, strongly coupled collective states where the plasma resembles a solid (e.g., forms quasi-crystalline structures).

In addition to the above activities there have been continuing investigations of industrially relevant plasmas such as *arcs*, *plasma torches*, and *laser plasmas*. In particular, approximately 40% of the steel manufactured in the United States is recycled in huge electric arc furnaces capable of melting over 100 tons of scrap steel in a few minutes. Plasma displays are used for flat-panel televisions and of course there are naturally occurring terrestrial plasmas such as *lightning*.

1.3 Plasma parameters

Three fundamental parameters¹ characterize a plasma:

1. the particle density n (measured in particles per cubic meter),
2. the temperature T of each species (usually measured in eV, where 1 eV = 11 605 K),
3. the steady-state magnetic field B (measured in Tesla).

A host of subsidiary parameters (e.g., Debye length, Larmor radius, plasma frequency, cyclotron frequency, thermal velocity) can be derived from these three fundamental parameters. For partially ionized plasmas, the fractional ionization and cross-sections of neutrals are also important.

¹ In older plasma literature, density and magnetic fields are often expressed in cgs units, i.e., densities are given in particles per cubic centimeter, and magnetic fields are given in Gauss. Since the 1990s there has been general agreement to use SI units when possible. SI units have the distinct advantage that electrical units are in terms of familiar quantities such as amps, volts, and ohms and so a model prediction in SI units can much more easily be compared to the results of an experiment than a prediction given in cgs units.

1.4 Examples of plasmas

1.4.1 *Non-fusion terrestrial plasmas*

It takes considerable resources and skill to make a hot, fully ionized plasma and so, except for the specialized fusion plasmas, most terrestrial plasmas (e.g., arcs, neon signs, fluorescent lamps, processing plasmas, welding arcs, and lightning) have electron temperatures of a few eV and, for reasons given later, have ion temperatures that are colder, often at room temperature. These “everyday” plasmas usually have no imposed steady-state magnetic field and do not produce significant self-magnetic fields. Typically, these plasmas are weakly ionized and dominated by collisional and radiative processes. Densities in these plasmas range from 10^{14} to 10^{22} m^{-3} (for comparison, the density of air at STP is $2.7 \times 10^{25} \text{ m}^{-3}$).

1.4.2 *Fusion-grade terrestrial plasmas*

Using carefully designed, expensive, and often large plasma confinement systems together with high heating power and obsessive attention to purity, fusion researchers have succeeded in creating fully ionized hydrogen or deuterium plasmas which attain temperatures ranging from tens of eV to *tens of thousands* of eV. In typical magnetic confinement devices (e.g., tokamaks, stellarators, reversed field pinches, mirror devices) an externally produced 1–10 T magnetic field of carefully chosen geometry is imposed on the plasma. Magnetic confinement devices generally have densities in the range 10^{19} – 10^{21} m^{-3} . Plasmas used in inertial fusion are much more dense; the goal is to attain for a brief instant densities one or two orders of magnitude larger than solid density ($\sim 10^{28} \text{ m}^{-3}$).

1.4.3 *Space plasmas*

The parameters of these plasmas cover an enormous range. For example, the density of space plasmas varies from 10^6 m^{-3} in interstellar space to 10^{20} m^{-3} in the solar atmosphere. Most of the astrophysical plasmas that have been investigated have temperatures in the range of 1–100 eV and these plasmas are usually fully ionized.

1.5 Logical framework of plasma physics

Plasmas are complex and exist in a wide variety of situations differing by many orders of magnitude. An important situation where plasmas do not normally exist is ordinary human experience. Consequently, people do not have the sort of intuition for plasma behavior that they have for solids, liquids, or gases. Although plasma behavior seems non- or counter-intuitive at first, with suitable effort a good intuition for plasma behavior can be developed. This intuition can

be helpful for making initial predictions about plasma behavior in a new situation, because plasmas have the remarkable property of being extremely *scalable*; i.e., the same qualitative phenomena often occur in plasmas differing by many orders of magnitude. Plasma physics is usually not a precise science. It is rather a web of overlapping points of view, each modeling a limited range of behavior. Understanding of plasmas is developed by studying these various points of view, all the while keeping in mind the linkages between the points of view.

Plasma dynamics is determined by the *self-consistent interaction between electromagnetic fields and statistically large numbers of charged particles* as shown schematically in Fig. 1.1. In principle, the time evolution of a plasma can be calculated as follows:

1. given the trajectory $\mathbf{x}_j(t)$ and velocity $\mathbf{v}_j(t)$ of each and every particle j , the electric field $\mathbf{E}(\mathbf{x}, t)$ and magnetic field $\mathbf{B}(\mathbf{x}, t)$ can be evaluated using Maxwell's equations and simultaneously;
2. given the instantaneous electric and magnetic fields $\mathbf{E}(\mathbf{x}, t)$ and $\mathbf{B}(\mathbf{x}, t)$, the forces on each and every particle j can be evaluated using the Lorentz equation and then used to update the trajectory $\mathbf{x}_j(t)$ and velocity $\mathbf{v}_j(t)$ of each particle.

While this approach is conceptually easy to understand, it is normally impractical to implement because of the extremely large number of particles and, to a lesser extent, because of the complexity of the electromagnetic field. To gain a practical understanding, we therefore do not attempt to evaluate the entire complex behavior all at once but, instead, study plasmas by considering specific phenomena. For each phenomenon under immediate consideration, appropriate simplifying approximations are made, leading to a more tractable problem and hopefully revealing the essence of what is going on. A situation where a certain set of approximations is valid and provides a self-consistent description is called a regime. There are a number of general categories of simplifying approximations, namely:

1. *Approximations involving the electromagnetic field:*
 - (a) assuming the magnetic field is zero (unmagnetized plasma);
 - (b) assuming there are no inductive electric fields (electrostatic approximation);

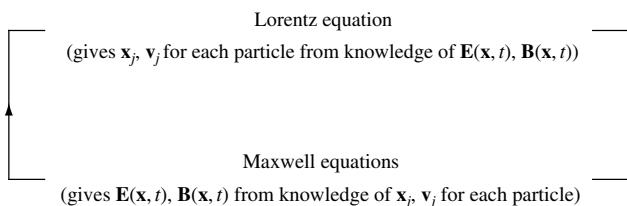


Fig. 1.1 Interrelation between Maxwell's equations and the Lorentz equation.

- (c) neglecting the displacement current in Ampère's law (suitable for phenomena having characteristic velocities much slower than the speed of light);
- (d) assuming that all magnetic fields are produced by conductors external to the plasma;
- (e) various assumptions regarding geometric symmetry (e.g., spatially uniform, uniform in a particular direction, azimuthally symmetric about an axis).

2. *Approximations involving the particle description:*

- (a) Averaging of the Lorentz force over some sub-group of particles:
 - i. Vlasov theory: average over all particles of a given species (electrons or ions) having the same velocity at a given location and characterize the plasma using the distribution function $f_\sigma(\mathbf{x}, \mathbf{v}, t)$, which gives the density of particles of species σ having velocity $\mathbf{v}$ at position $\mathbf{x}$ at time t ;
 - ii. Two-fluid theory: average velocities over all particles of a given species at a given location and characterize the plasma using the species density $n_\sigma(\mathbf{x}, t)$, mean velocity $\mathbf{u}_\sigma(\mathbf{x}, t)$, and pressure $P_\sigma(\mathbf{x}, t)$ defined relative to the species mean velocity.
 - iii. Magnetohydrodynamic theory: average momentum over all particles of all species and characterize the plasma using the center-of-mass density $\rho(\mathbf{x}, t)$, center-of-mass velocity $\mathbf{U}(\mathbf{x}, t)$, and pressure $P(\mathbf{x}, t)$ defined relative to the center-of-mass velocity.
- (b) Assumptions about time (e.g., assume the phenomenon under consideration is fast or slow compared to some characteristic frequency of the particles such as the cyclotron frequency).
- (c) Assumptions about space (e.g., assume the scale length of the phenomenon under consideration is large or small compared to some characteristic plasma length such as the cyclotron radius).
- (d) Assumptions about velocity (e.g., assume the phenomenon under consideration is fast or slow compared to the thermal velocity $v_{T\sigma}$ of a particular species σ).

The large number of possible permutations and combinations that can be constructed from the above list means that there will be a large number of regimes. Since developing an intuitive understanding requires making approximations of the sort listed above and since these approximations lack an obvious hierarchy, it is not clear where to begin. In fact, as sketched in Fig. 1.2, the models for particle motion (Vlasov, two-fluid, MHD) involve a circular argument. Wherever we start on this circle, we are always forced to take at least one new concept on trust and hope that its validity will be established later. The reader is encouraged to refer to Fig. 1.2 as its various components are examined so that the logic of this circle will eventually become clear.

Because the argument is circular, the starting point is at the author's discretion, and for good (but not overwhelming) reasons, this author has decided that the optimum starting point on Fig. 1.2 is the subject of *Debye shielding*. Debye concepts, the Rutherford model for how charged particles scatter from each other,

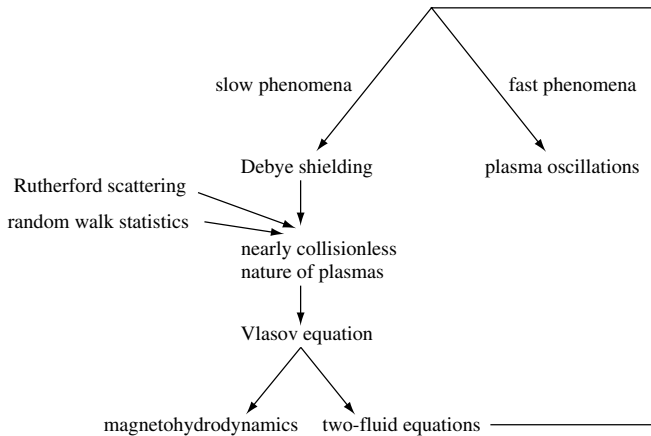


Fig. 1.2 Hierarchy of models of plasmas showing circular nature of logic.

and some elementary statistics will be combined to construct an argument showing that plasmas are *weakly collisional*. We will then discuss *phase-space* concepts and introduce the *Vlasov equation* for the phase-space density. Averages of the Vlasov equation will provide *two-fluid equations* and also the *magnetohydrodynamic* (MHD) equations. Having established this framework, we will then return to study features of these points of view in more detail, often tying up loose ends that occurred in our initial derivation of the framework. Somewhat separate from the study of Vlasov, two-fluid, and MHD equations (which all attempt to give a self-consistent picture of the plasma) is the study of *single particle orbits in prescribed fields*. This provides useful intuition on the behavior of a typical particle in a plasma, and can provide important inputs or constraints for the self-consistent theories.

1.6 Debye shielding

We begin our study of plasmas by examining Debye shielding, a concept originating from the theory of liquid electrolytes (Debye and Hückel 1923). Consider a finite-temperature plasma consisting of a statistically large number of electrons and ions and assume that the ion and electron densities are initially equal and spatially uniform. As will be seen later, the ions and electrons need not be in thermal equilibrium with each other, and so the ions and electrons will be allowed to have separate temperatures denoted by T_i , T_e .

Since the ions and electrons have random thermal motion, thermally induced perturbations about the equilibrium will cause small, transient spatial variations

of the electrostatic potential ϕ . In the spirit of circular argument the following assumptions are now invoked without proof:

1. The plasma is assumed to be nearly collisionless so that collisions between particles may be neglected to first approximation.
2. Each species, denoted as σ , may be considered as a “fluid” having a density n_σ , a temperature T_σ , a pressure $P_\sigma = n_\sigma \kappa T_\sigma$ (κ is Boltzmann’s constant), and a mean velocity $\mathbf{u}_\sigma$ so that the collisionless equation of motion for each fluid is

$$m_\sigma \frac{d\mathbf{u}_\sigma}{dt} = q_\sigma \mathbf{E} - \frac{1}{n_\sigma} \nabla P_\sigma, \quad (1.1)$$

where m_σ is the particle mass, q_σ is the charge of a particle, and $\mathbf{E}$ is the electric field.

Now consider a perturbation with a sufficiently *slow* time dependence to allow the following assumptions:

1. The inertial term $\sim d/dt$ on the left-hand side of Eq. (1.1) is negligible and may be dropped.
2. Inductive electric fields are negligible so the electric field is almost entirely electrostatic, i.e., $\mathbf{E} \sim -\nabla\phi$.
3. All temperature gradients are smeared out by thermal particle motion so that the temperature of each species is spatially uniform.
4. The plasma remains in thermal equilibrium throughout the perturbation (i.e., it can always be characterized by a temperature).

Invoking these approximations, Eq. (1.1) reduces to

$$0 \approx -n_\sigma q_e \nabla\phi - \kappa T_\sigma \nabla n_\sigma, \quad (1.2)$$

a simple balance between the force due to the electrostatic electric field and the force due to the isothermal pressure gradient. Equation (1.2) is readily solved to give the *Boltzmann relation*

$$n_\sigma = n_{\sigma 0} \exp(-q_\sigma \phi / \kappa T_\sigma), \quad (1.3)$$

where $n_{\sigma 0}$ is a constant. It is important to emphasize that the Boltzmann relation results from the assumption that the perturbation is *very slow*; if this is not the case, then inertial effects, inductive electric fields, or temperature gradient effects will cause the plasma to have a completely different behavior from the Boltzmann relation. Situations exist where this “slowness” assumption is valid for electron dynamics but not for ion dynamics, in which case the Boltzmann condition will apply only to the electrons but not to the ions (the converse situation does not normally occur because ions, being heavier, are always more sluggish than electrons and so it is only possible for a phenomenon to appear slow to electrons but not to ions).

Let us now imagine slowly inserting a single additional particle (so-called “test” particle) with charge q_T into an initially unperturbed, spatially uniform, neutral plasma. To keep the algebra simple, we define the origin of our coordinate system to be at the location of the test particle. Before insertion of the test particle, the plasma potential was $\phi = 0$ everywhere because the ion and electron densities were spatially uniform and equal, but now the ions and electrons will be perturbed because of their interaction with the test particle. Particles having the same polarity as q_T will be slightly repelled whereas particles of opposite polarity will be slightly attracted. The slight displacements resulting from these repulsions and attractions will result in a small, but finite, potential in the plasma. This potential will be the superposition of the test particle’s own potential and the potential of the plasma particles that have moved slightly in response to the test particle.

This slight displacement of plasma particles is called *shielding* or *screening* of the test particle because the displacement tends to reduce the effectiveness of the test particle field. To see this, suppose the test particle is a positively charged ion. When immersed in the plasma it will attract nearby electrons and repel nearby ions; the net result is an effectively negative charge cloud surrounding the test particle. An observer located far from the test particle and its surrounding cloud would see the combined potential of the test particle and its associated cloud. Because the cloud has the opposite polarity to the test particle, the cloud potential will partially cancel (i.e., *shield* or *screen*) the test particle potential.

Screening is calculated using Poisson’s equation with the source terms being the test particle and its associated cloud. The cloud contribution is determined using the Boltzmann relation for the particles that participate in the screening. This is a “self-consistent” calculation for the potential because the shielding cloud is affected by its self-potential.

Thus, Poisson’s equation becomes

$$\nabla^2 \phi = -\frac{1}{\epsilon_0} \left[q_T \delta(\mathbf{r}) + \sum_{\sigma} n_{\sigma}(\mathbf{r}) q_{\sigma} \right], \quad (1.4)$$

where the term $q_T \delta(\mathbf{r})$ on the right-hand side represents the charge density due to the test particle and the term $\sum_{\sigma} n_{\sigma}(\mathbf{r}) q_{\sigma}$ represents the charge density of all plasma particles that participate in the screening (i.e., everything except the test particle). Before the test particle was inserted $\sum_{\sigma=i,e} n_{\sigma}(\mathbf{r}) q_{\sigma}$ vanished because the plasma was assumed to be initially neutral.

Since the test particle was inserted slowly, the plasma response will be Boltzmann-like and we may substitute for $n_{\sigma}(\mathbf{r})$ using Eq. (1.3). Furthermore, because the perturbation due to a single test particle is infinitesimal, we can

safely assume that $|q_\sigma \phi| \ll \kappa T_\sigma$, in which case Eq. (1.3) becomes simply $n_\sigma/n_{\sigma 0} \approx 1 - q_\sigma \phi / \kappa T_\sigma$ and so Eq. (1.4) becomes

$$\nabla^2 \phi = -\frac{1}{\varepsilon_0} \left[q_T \delta(\mathbf{r}) + \left(1 - \frac{q_e \phi}{\kappa T_e} \right) n_{e0} q_e + \left(1 - \frac{q_i \phi}{\kappa T_i} \right) n_{i0} q_i \right]. \quad (1.5)$$

The assumption of initial neutrality means that $n_{e0} q_e + n_{i0} q_i = 0$, in which case Eq. (1.5) reduces to

$$\nabla^2 \phi - \frac{1}{\lambda_D^2} \phi = -\frac{q_T}{\varepsilon_0} \delta(\mathbf{r}), \quad (1.6)$$

where the effective Debye length is defined by

$$\frac{1}{\lambda_D^2} = \sum \frac{1}{\lambda_\sigma^2} \quad (1.7)$$

and the species Debye length λ_σ is

$$\lambda_\sigma^2 = \frac{\varepsilon_0 \kappa T_\sigma}{n_{\sigma 0} q_\sigma^2}. \quad (1.8)$$

The second term on the left-hand side of Eq. (1.6) is just the negative of the shielding cloud charge density. The summation in Eq. (1.7) is over all species that participate in the shielding. Since ions cannot move fast enough to keep up with an electron test charge, which would be moving at the nominal electron thermal velocity, the shielding of electrons is only by other electrons, whereas the shielding of ions is by both ions and electrons.

Equation (1.6) can be solved using standard mathematical techniques (cf. assignments) to give

$$\phi(\mathbf{r}) = \frac{q_T}{4\pi\varepsilon_0 r} e^{-r/\lambda_D}; \quad (1.9)$$

this is sometimes called the Yukawa potential. For $r \ll \lambda_D$ the potential $\phi(\mathbf{r})$ is identical to the potential of a test particle in vacuum, whereas for $r \gg \lambda_D$ the test charge is completely screened by its surrounding shielding cloud. The nominal radius of the shielding cloud is λ_D . Because the test particle is completely screened for $r \gg \lambda_D$, the total shielding cloud charge is equal in magnitude to the charge on the test particle and opposite in sign. This test-particle/shielding-cloud analysis makes sense only if there is a macroscopically large number of plasma particles in the shielding cloud; i.e., the analysis makes sense only if $4\pi n_0 \lambda_D^3 / 3 \gg 1$. This will be seen later to be the condition for the plasma to be nearly collisionless and so validate assumption 1 at the top of p. 8.

In order for shielding to be a relevant issue, the Debye length must be small compared to the overall dimensions of the plasma, because otherwise no point in the plasma could be outside the shielding cloud. Finally, it should be realized

that *any* particle could have been construed as being “the” test particle and so we conclude that the time-averaged effective potential of any selected particle in the plasma is given by Eq. (1.9). (From a statistical point of view, a selected particle is no longer assumed to have a random thermal velocity, in which case its effective potential results from the summation of the non-random, static potential associated with its own charge and a screening potential, which is the time-average of a rapidly changing potential associated with the fleeting, random thermal motions of all other particles.)

1.7 Quasi-neutrality

The Debye shielding analysis assumed that the plasma was initially neutral, i.e., that the initial electron and ion densities were equal. We now demonstrate that if the Debye length is a microscopic length, then it is indeed an excellent assumption that plasmas remain extremely close to neutrality, while not being exactly neutral. It is found that the electrostatic electric field associated with any reasonable configuration is easily produced by having only a tiny deviation from perfect neutrality. This tendency to be *quasi-neutral* occurs because a conventional plasma does not have sufficient internal energy to become substantially non-neutral over distances exceeding a Debye length (there do exist non-neutral plasmas that violate this concept, but these involve rotation of plasma in a background magnetic field, which effectively plays the neutralizing role of ions in a conventional plasma).

To prove the assertion that plasmas tend to be quasi-neutral, we consider an initially neutral plasma with temperature T and calculate the largest radius sphere that could spontaneously become depleted of electrons due to thermal fluctuations. Let $r_{\max}$ be the radius of this presumed sphere. Complete depletion (i.e., maximum non-neutrality) would occur if a random thermal fluctuation caused all the electrons originally in the sphere to vacate the volume of the sphere and move to its surface. The electrons would have to come to rest on the surface of the presumed sphere because if they did not, they would still have available kinetic energy, which could then be used to move out towards an even larger radius, violating the assumption that the sphere was the largest radius sphere that could become fully depleted of electrons. This situation is of course extremely artificial and likely to be so rare as to be essentially negligible because it requires all the electrons to be moving radially relative to some origin. In reality, the electrons would be moving in random directions.

When the electrons exit the sphere they leave behind an equal number of ions. The remnant ions produce a radial electric field, which pulls the electrons back towards the center of the sphere. One way of calculating the energy stored in this system is to calculate the work done by the electrons as they leave the sphere

and collect on the surface, but a simpler way is to calculate the energy stored in the electrostatic electric field produced by the ions remaining in the sphere. This electrostatic energy did not exist when the electrons were initially in the sphere and balanced the ion charge and so it must be equivalent to the work done by the electrons on leaving the sphere.

The energy density of an electric field is $\epsilon_0 E^2/2$ and, because of the spherical symmetry assumed here, the electric field produced by the remnant ions must be in the radial direction. The ion charge in a sphere of radius r is $Q = 4\pi n e r^3/3$ and so, after all the electrons have vacated the sphere, the electric field at radius r is therefore $E_r = Q/4\pi\epsilon_0 r^2 = n e r/3\epsilon_0$. The electrostatic field energy in the ion-filled, electron-depleted sphere of radius $r_{\max}$ is thus

$$W = \int_0^{r_{\max}} \frac{\epsilon_0 E_r^2}{2} 4\pi r^2 dr = \pi r_{\max}^5 \frac{2n_e^2 e^2}{45\epsilon_0}. \quad (1.10)$$

Equating this potential energy to the initial electron thermal kinetic energy, W_{kinetic} , gives

$$\pi r_{\max}^5 \frac{2n_e^2 e^2}{45\epsilon_0} = \frac{3}{2} n \kappa T \times \frac{4}{3} \pi r_{\max}^3, \quad (1.11)$$

which may be solved to give

$$r_{\max}^2 = 45 \frac{\epsilon_0 \kappa T}{n_e e^2} \quad (1.12)$$

so that $r_{\max} \simeq 7\lambda_D$.

Thus, the largest spherical volume that could spontaneously become depleted of electrons has a radius of a few Debye lengths. Since the probability of all electrons in a sphere simultaneously having radially outwards velocities at some instant is essentially zero, this electron depletion of a sphere is extremely unlikely and so regions of non-neutrality will be much smaller than given by Eq. (1.12). We conclude that *plasma is quasi-neutral over scale lengths much larger than the Debye length*. When a biased electrode such as a wire probe is inserted into a plasma, the plasma screens the field due to the potential on the electrode in the same way that the test charge potential was screened. The screening region is called the *sheath*, which is a region of non-neutrality having an extent of the order of a Debye length.

1.8 Small- vs. large-angle collisions in plasmas

We now consider what happens to the momentum and energy of a test particle of charge q_T and mass m_T that is injected with velocity $\mathbf{v}_T$ into a plasma. This test particle will make a sequence of random collisions with the plasma particles

(called “field” particles and denoted by subscript F); these collisions will alter both the momentum and the energy of the test particle. A collision is characterized by a deflection (also called scattering) of the test particle by an angle θ from its original direction of motion. The scattering angle θ depends on a number of parameters, including the test and target particle masses, the speed of the test particle relative to the target particle, and whether the collision is a bull’s-eye collision or merely a grazing collision. Determining this functional dependence is called the Rutherford scattering problem, a standard problem in classical mechanics.

Solution of the Rutherford scattering problem in the center-of-mass frame shows (Assignment 1, this chapter) that the scattering angle θ is given by

$$\tan\left(\frac{\theta}{2}\right) = \frac{q_T q_F}{4\pi\epsilon_0 b \mu v_0^2} \sim \frac{\text{Coulomb interaction energy}}{\text{kinetic energy}}, \quad (1.13)$$

where $\mu^{-1} = m_T^{-1} + m_F^{-1}$ is the reduced mass, b is the impact parameter shown in Fig. 1.3, and v_0 is the initial relative velocity. It is useful to separate scattering events (i.e., collisions) into two approximate categories, namely (1) large-angle collisions where $\pi/2 \leq \theta \leq \pi$ and (2) small-angle (grazing) collisions where $\theta \ll \pi/2$.

Let us denote $b_{\pi/2}$ as the impact parameter for 90° collisions; from Eq. (1.13) this is

$$b_{\pi/2} = \frac{q_T q_F}{4\pi\epsilon_0 \mu v_0^2} \quad (1.14)$$

and corresponds to the radius of the inner (small) shaded circle in Fig. 1.3. Large-angle scatterings will occur if the test particle is incident anywhere within this

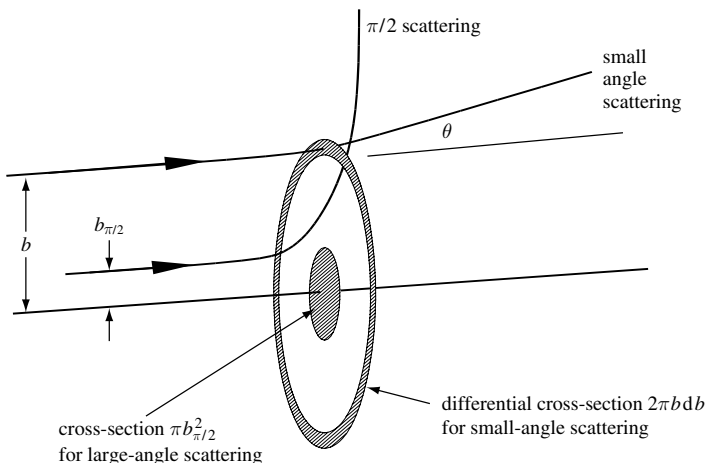


Fig. 1.3 Differential scattering cross-sections for large and small deflections.

circle and so the total cross-section for *all* large-angle collisions is

$$\begin{aligned}\sigma_{\text{large}} &\approx \pi b_{\pi/2}^2 \\ &= \pi \left(\frac{q_T q_F}{4\pi\epsilon_0 \mu v_0^2} \right)^2.\end{aligned}\quad (1.15)$$

Grazing (small-angle) collisions occur when the test particle impinges *outside* the shaded circle and so *occur much more frequently* than large-angle collisions. Although each grazing collision does not deflect the test particle very much, there are far more grazing collisions than large-angle collisions and so it is important to compare the *cumulative* effect of grazing collisions with the *cumulative* effect of large-angle collisions.

To make matters even more complicated, the effective cross-section of grazing collisions depends on impact parameter, since the larger b is, the smaller the deflection. To take this weighting of impact parameters into account, the area outside the shaded circle is subdivided into a set of concentric annuli, called differential cross-sections. If the test particle impinges on the differential cross-section having radius between b and $b+db$, then the test particle will be deflected by an angle lying between $\theta(b)$ and $\theta(b+db)$ as determined by Eq. (1.13). The area of the differential cross-section is $2\pi b db$, which is therefore the effective cross-section for scattering between $\theta(b)$ and $\theta(b+db)$. Because the azimuthal angle about the direction of incidence is random, the simple average of N small-angle deflections vanishes, i.e., $N^{-1} \sum_{i=1}^N \theta_i = 0$, where θ_i is the deflection due to the i th collision and N is a large number.

Random walk statistics must therefore be used to describe the cumulative effect of a large number of small-angle deflections and so we will use the *square* of the deflection angle, i.e., θ_i^2 , as the quantity for comparing the cumulative effects of small (grazing) and large-angle collisions. Thus, scattering (i.e., a sequence of random deflections) is a diffusive process.

To compare the respective cumulative effects of grazing and large-angle collisions, we calculate how many random small-angle deflections must occur to be equivalent to a single large-angle deflection (i.e., $\theta_{\text{large}}^2 \approx 1$); here we pick the nominal value of the large-angle deflection to be 1 radian. In other words, we ask what must N be in order to have $\sum_{i=1}^N \theta_i^2 \approx 1$, where each θ_i represents an individual small-angle deflection (sometimes called a “scattering event”). Equivalently, we may ask what time t do we have to wait for the cumulative effect of the grazing collisions on a test particle to give an effective deflection equivalent to a single large-angle deflection.

To calculate this, let us imagine we are “sitting” on the test particle. In this test particle frame the field particles approach the test particle with the velocity

v_{rel} , and so the apparent flux of field particles is $\Gamma = n_F v_{rel}$, where v_{rel} is the relative velocity between the test and field particles. The number of small-angle scattering events in time t for impact parameters between b and $b + db$ is $\Gamma t 2\pi b db$ and so the time required for the cumulative effect of small-angle collisions to be equivalent to a large-angle collision is given by

$$1 \approx \sum_{i=1}^N \theta_i^2 = \Gamma t \int 2\pi b db [\theta(b)]^2. \quad (1.16)$$

The definitions of scattering theory show (see Assignment 9) that $\sigma\Gamma = t^{-1}$, where σ is the cross-section for an event and t is the time one has to wait for the event to occur. Substituting for Γt in Eq. (1.16) gives the cross-section σ^* for the cumulative effect of grazing collisions to be equivalent to a single large-angle scattering event,

$$\sigma^* = \int 2\pi b db [\theta(b)]^2. \quad (1.17)$$

The appropriate lower limit for the integral in Eq. (1.17) is $b_{\pi/2}$, since impact parameters smaller than this value produce large-angle collisions. What should the upper limit of the integral be? We recall from our Debye discussion that the field of the scattering center is *screened out* for distances greater than λ_D . Hence, small-angle collisions occur *only* for impact parameters in the range $b_{\pi/2} < b < \lambda_D$ because the scattering potential is non-existent for distances larger than λ_D .

For small-angle collisions, Eq. (1.13) gives

$$\theta(b) = \frac{q_T q_F}{2\pi\epsilon_0 \mu v_0^2 b} \quad (1.18)$$

so Eq. (1.17) becomes

$$\sigma^* = \int_{b_{\pi/2}}^{\lambda_D} 2\pi b db \left(\frac{q_T q_F}{2\pi\epsilon_0 \mu v_0^2 b} \right)^2 \quad (1.19)$$

or

$$\sigma^* = 8 \ln \left(\frac{\lambda_D}{b_{\pi/2}} \right) \sigma_{\text{large}}. \quad (1.20)$$

Thus, the cross-section σ^* will significantly exceed σ_{large} if $\lambda_D/b_{\pi/2} \gg 1$. Since $b_{\pi/2} = 1/2n\lambda_D^2$, the condition $\lambda_D \gg b_{\pi/2}$ is equivalent to $n\lambda_D^3 \gg 1$, which is just the criterion for there to be a large number of particles in a sphere having a radius λ_D (a so-called *Debye sphere*). This was the condition for the Debye shielding cloud argument to make sense. We conclude that the criterion for an ionized gas to behave as a plasma (i.e., Debye shielding is important and grazing collisions dominate large-angle collisions) is the condition $n\lambda_D^3 \gg 1$. For most

plasmas $n\lambda_D^3$ is a large number with natural logarithm of order 10; typically, when making rough estimates of σ^* , one uses $\ln(\lambda_D/b_{\pi/2}) \approx 10$. The reader may have developed a concern about the seeming arbitrary nature of the choice of $b_{\pi/2}$ as the “dividing line” between large-angle and grazing collisions. This arbitrariness is of no consequence since the logarithmic dependence means that any other choice having the same order of magnitude for the “dividing line” would give essentially the same result.

By substituting for $b_{\pi/2}$, the cross-section can be rewritten as

$$\sigma^* = \frac{1}{2\pi} \left(\frac{q_T q_F}{\epsilon_0 \mu v_0^2} \right)^2 \ln \left(\frac{\lambda_D}{b_{\pi/2}} \right). \quad (1.21)$$

Thus, σ^* decreases approximately as the fourth power of the relative velocity. In a hot plasma where v_0 is large, σ^* will be *very small* and so scattering by Coulomb collisions is often much less important than other phenomena. A useful way to decide whether Coulomb collisions are important is to compare the collision frequency $\nu = \sigma^* n v$ with the frequency of other effects, or equivalently the mean free path of collisions, $l_{mfp} = 1/\sigma^* n$, with the characteristic length of other effects. If the collision frequency is small, or the mean free path is large (in comparison to other effects), collisions may be neglected to first approximation, in which case the plasma under consideration is called a collisionless or “ideal” plasma. The effective Coulomb cross-section σ^* and its related parameters ν and l_{mfp} can be used to evaluate transport properties such as electrical resistivity, mobility, and diffusion.

1.9 Electron and ion collision frequencies

One of the fundamental physical constants influencing plasma behavior is the ion to electron mass ratio. The large value of this ratio often causes electrons and ions to experience qualitatively distinct dynamics. In some situations, one species may determine the essential character of a particular plasma behavior while the other species has little or no effect. Let us now examine how mass ratio affects:

1. momentum change (scattering) of a given incident particle due to collision between
 - (a) like particles (i.e., electron-electron or ion-ion collisions, denoted *ee* or *ii*),
 - (b) unlike particles (i.e., electrons scattering from ions denoted *ei* or ions scattering from electrons denoted *ie*);
2. kinetic energy change (scattering) of a given incident particle due to collisions between like or unlike particles.

Momentum scattering is characterized by the time required for collisions to deflect the incident particle by an angle $\pi/2$ from its initial direction or, more

commonly, by the inverse of this time, called the collision frequency. The momentum scattering collision frequencies are denoted as $\nu_{ee}, \nu_{ii}, \nu_{ei}, \nu_{ie}$ for the various possible interactions between species and the corresponding times as τ_{ee} , etc. Energy scattering is characterized by the time required for an incident particle to transfer all its kinetic energy to the target particle. Energy transfer collision frequencies are denoted respectively by $\nu_{Eee}, \nu_{Eii}, \nu_{Eei}, \nu_{Eie}$.

We now show that these frequencies separate into categories having *three distinct orders of magnitude* with relative scalings $1 : (m_i/m_e)^{1/2} : m_i/m_e$. In order to estimate the orders of magnitude of the collision frequencies we assume the incident particle is “typical” for its species and so take its incident velocity to be the species thermal velocity, $v_{T\sigma} = (2\kappa T_\sigma/m_\sigma)^{1/2}$. While this is reasonable for a rough estimate, it should be realized that, because of the v^{-4} dependence in σ^* , a more careful averaging over all particles in the thermal distribution will differ. This careful averaging is quite intricate and will be deferred to Chapter 13.

We normalize all collision frequencies to ν_{ee} , and for further simplification assume that the ion and electron temperatures are of the same order of magnitude. First consider ν_{ei} : the reduced mass for ei collisions is the same as for ee collisions (except for a factor of 2, which we neglect) and the relative velocity is the same – hence, we conclude that $\nu_{ei} \sim \nu_{ee}$. Now consider ν_{ii} : because the temperatures were assumed equal, $\sigma_{ii}^* \approx \sigma_{ee}^*$ and so the collision frequencies will differ only because of the different velocities in the expression $\nu = n\sigma v$. The ion thermal velocity is lower by an amount $(m_e/m_i)^{1/2}$, giving $\nu_{ii} \approx (m_e/m_i)^{1/2} \nu_{ee}$.

Care is required when calculating ν_{ie} . Strictly speaking, this calculation should be done in the center-of-mass frame and then transformed back to the lab frame, but an easy way to estimate ν_{ie} using lab-frame calculations is to note that momentum is conserved in a collision so that in the lab frame $m_i \Delta \mathbf{v}_i = -m_e \Delta \mathbf{v}_e$, where Δ means the change in a quantity as a result of the collision. If the collision of an ion head-on with a stationary electron is taken as an example, then the electron bounces off forward with twice the ion’s velocity (corresponding to a specular reflection of the electron in a frame where the ion is stationary); this gives $\Delta \mathbf{v}_e = 2\mathbf{v}_i$ and $|\Delta \mathbf{v}_i|/|\mathbf{v}_i| = 2m_e/m_i$. Thus, in order to have $|\Delta \mathbf{v}_i|/|\mathbf{v}_i|$ of order unity, it is necessary to have m_i/m_e head-on collisions of an ion with electrons, whereas in order to have $|\Delta \mathbf{v}_e|/|\mathbf{v}_e|$ of order unity it is only necessary to have one collision of an electron with an ion. Hence, $\nu_{ie} \sim (m_e/m_i) \nu_{ee}$.

Now consider energy changes in collisions. If a moving electron makes a head-on collision with an electron at rest, then the incident electron stops (loses all its momentum and energy), while the originally stationary electron flies off with the same momentum and energy that the incident electron had. A similar picture holds for an ion hitting an ion. Thus, like-particle collisions transfer energy at the same rate as momentum, so $\nu_{Eee} \sim \nu_{ee}$ and $\nu_{Eii} \sim \nu_{ii}$.

Table 1.1

~ 1	$\sim (m_e/m_i)^{1/2}$	$\sim m_e/m_i$
ν_{ee}	ν_{ii}	ν_{ie}
ν_{ei}	ν_{Eii}	ν_{Eei}
ν_{Eee}		ν_{Eie}

Interspecies collisions are more complicated. Consider an electron hitting a stationary ion head-on. Because the ion is massive, it barely recoils and the electron reflects with a velocity nearly equal in magnitude to its incident velocity. Thus, the change in electron momentum is $-2m_e\mathbf{v}_e$. From conservation of momentum, the momentum of the recoiling ion must be $m_i\mathbf{v}_i = 2m_e\mathbf{v}_e$. The energy transferred to the ion in this collision is $m_iv_i^2/2 = 4(m_e/m_i)m_ev_e^2/2$. Thus, an electron has to make $\sim m_i/m_e$ such collisions in order to transfer all its energy to ions. Hence, $\nu_{Eei} = (m_e/m_i)\nu_{ee}$.

Similarly, if an incident ion hits an electron at rest the electron will fly off with twice the incident ion velocity (in the center-of-mass frame, the electron is reflecting from the ion). The electron gains energy $m_ev_i^2/2$ so that, again, $\sim m_i/m_e$ collisions are required for the ion to transfer all its energy to electrons. The orders of magnitudes of collision frequencies are summarized in Table 1.1.

Although collisions are typically unimportant for fast, transient processes, they may eventually determine many properties of a given plasma. The wide disparity of collision frequencies shows that one has to be careful when determining which collisional process is relevant to a given phenomenon. Perhaps the best way to illustrate how collisions must be considered is by an example, such as the following:

Suppose half the electrons in a plasma initially have a directed velocity $\mathbf{v}_0$ while the other half of the electrons and all the ions are initially at rest. This may be thought of as a high-density beam of electrons passing through a cold plasma. On the fast (i.e., ν_{ee}) time scale the beam electrons will:

- (i) Collide with the stationary electrons and share their momentum and energy so that after a time of order ν_{ee}^{-1} the beam will become indistinguishable from the background electrons. Since momentum must be conserved, the combined electrons will have a mean velocity $\mathbf{v}_0/2$.
- (ii) Collide with the stationary ions, which will act as nearly fixed scattering centers so that the beam electrons will scatter in direction but not transfer significant energy to the ions.

Both of the above processes will randomize the velocity distribution of the electrons until this distribution becomes Maxwellian (the maximum entropy

distribution); the Maxwellian will be centered about the average velocity discussed in (i) above.

On the very slow ν_{Eei} time scale (down by a factor m_i/m_e), the electrons will transfer momentum to the ions, so on this time scale the electrons will share their momentum with the ions, in which case the electrons will slow down and the ions will speed up until the average electron velocity and the average ion velocity become identical. Similarly, the electrons will share energy with the ions, in which case the ions will heat up while the electrons will cool.

If, instead, a beam of ions were injected into the plasma, the ion beam would thermalize and share momentum with the background ions on the intermediate ν_{ii} time scale, and then only share momentum and energy with the electrons on the very slow ν_{Eie} time scale.

This collisional sharing of momentum and energy and thermalization of velocity distribution functions to make Maxwellians is the process by which thermodynamic equilibrium is achieved. Collision frequencies vary as $T^{-3/2}$ and so, for hot plasmas, collision processes are often slower than many other phenomena. Since collisions are the means by which thermodynamic equilibrium is achieved, plasmas are typically *not in thermodynamic equilibrium*, although some components of the plasma may be in a partial equilibrium (for example, the electrons may be in thermal equilibrium with each other but not with the ions). Hence, thermodynamically based descriptions of the plasma are often inappropriate. It is not unusual, for example, to have a plasma where the electron and ion temperatures differ by more than an order of magnitude. This can occur when one species or the other has been subject to heating and the plasma lifetime is shorter than the interspecies energy equilibration time, $\sim \nu_{Eei}^{-1}$.

1.10 Collisions with neutrals

If a plasma is weakly ionized then collisions with neutrals must be considered. These collisions differ fundamentally from collisions between charged particles because now the interaction forces are short-range (unlike the long-range Coulomb interaction) and so the neutral can be considered simply as a hard body with cross-section of the order of its actual geometrical size. All atoms have radii of the order of 10^{-10} m so the typical neutral cross-section is $\sigma_{\text{neut}} \sim 3 \times 10^{-20}$ m². When a particle hits a neutral it can simply scatter with no change in the internal energy of the neutral; this is called *elastic scattering*. It can also transfer energy to the structure of the neutral and so cause an internal change in the neutral; this is called *inelastic scattering*. Inelastic scattering includes ionization and excitation of atomic level transitions (with accompanying optical radiation).

Another process can occur when ions collide with neutrals – the incident ion can capture an electron from the neutral and become neutralized while simultaneously ionizing the original neutral. This process, called *charge exchange*, is used for producing energetic neutral beams. In this process a high-energy beam of ions is injected into a gas of neutrals, captures electrons, and exits as a high-energy beam of neutrals.

Because ions have approximately the same mass as neutrals, ions rapidly exchange energy with neutrals and tend to be in thermal equilibrium with the neutrals if the plasma is weakly ionized. As a consequence, ions are typically cold in weakly ionized plasmas, because the neutrals are in thermal equilibrium with the walls of the container.

1.11 Simple transport phenomena

1. Electrical resistivity. When a uniform electric field $\mathbf{E}$ exists in a plasma, the electrons and ions are accelerated in opposite directions creating a difference between the average velocities of the two species, i.e., a relative average velocity, $\mathbf{u}_{\text{rel}} = \mathbf{u}_e - \mathbf{u}_i$, of electrons relative to ions. The creation of this relative average velocity due to oppositely directed $\mathbf{E}$ field acceleration of electrons and ions competes with the simultaneous dissipation of the relative average velocity due to interspecies collisions (this dissipation of relative average velocity is known as “drag”). Balancing these competing forces on an average electron gives

$$0 = -e\mathbf{E} - v_{ei}m_e\mathbf{u}_{\text{rel}}, \quad (1.22)$$

since the drag force on the average electron is $v_{ei}m_e(\mathbf{u}_e - \mathbf{u}_i)$. However, since the electric current is $\mathbf{J} = -n_e e\mathbf{u}_{\text{rel}}$, Eq. (1.22) can be rewritten as

$$\mathbf{E} = \eta\mathbf{J}, \quad (1.23)$$

where

$$\eta = \frac{m_e v_{ei}}{n_e e^2} \quad (1.24)$$

is the plasma electrical resistivity. Substituting $v_{ei} = \sigma^* n_i v_{Te}$ and noting from quasi-neutrality that $Zn_i = n_e$, the plasma electrical resistivity is

$$\eta = \frac{Ze^2}{2\pi m_e \epsilon_0^2 v_{Te}^3} \ln \left(\frac{\lambda_D}{b_{\pi/2}} \right), \quad (1.25)$$

from which we see that resistivity is independent of density, proportional to $T_e^{-3/2}$, and also proportional to the ion charge Z . This expression for the resistivity is only approximate since we did not properly average over the electron velocity distribution (a more accurate expression, differing by a factor of order unity, will be derived in Chapter 13). Resistivity resulting from grazing collisions between electrons and ions as given by Eq. (1.25) is known as Spitzer resistivity (Spitzer and Harm 1953). It should

be emphasized that although this discussion assumes existence of a *uniform* electric field in the plasma, a uniform field will not exist in what naively appears to be the most obvious geometry, namely a plasma between two parallel plates charged to different potentials. This is because Debye shielding will concentrate virtually all the potential drop into thin sheaths adjacent to the electrodes, resulting in near-zero electric field inside the plasma. A practical way to obtain a uniform electric field is to create the field by induction so that there are no electrodes that can be screened out.

2. Diffusion and ambipolar diffusion. Standard random walk arguments show that particle diffusion coefficients scale as $D \sim (\Delta x)^2 / \tau$, where Δx is the characteristic step size in the random walk and τ is the time between steps. This can also be expressed as $D \sim v_T^2 / \nu$, where $\nu = \tau^{-1}$ is the collision frequency and $v_T = \Delta x / \tau = \nu \Delta x$ is the thermal velocity. Since the random step size for particle collisions is the mean free path and the time between steps is the inverse of the collision frequency, the electron diffusion coefficient in an unmagnetized plasma scales as

$$D_e = \nu_e l_{mfp,e}^2 = \frac{\kappa T_e}{m_e \nu_e}, \quad (1.26)$$

where $\nu_e = \nu_{ee} + \nu_{ei} \sim \nu_{ei}$ is the 90° scattering rate for electrons and $l_{mfp,e} = \sqrt{\kappa T_e / m_e \nu_e^2}$ is the electron mean free path. Similarly, the ion diffusion coefficient in an unmagnetized plasma is

$$D_i = \nu_i l_{mfp,i}^2 = \frac{\kappa T_i}{m_i \nu_i}, \quad (1.27)$$

where $\nu_i = \nu_{ii} + \nu_{ie} \sim \nu_{ie}$ is the effective ion collision frequency. The electron diffusion coefficient is typically much larger than the ion diffusion coefficient in an unmagnetized plasma (it is the other way around for diffusion across a magnetic field in a magnetized plasma where the step size is the Larmor radius). However, if the electrons in an unmagnetized plasma did in fact diffuse across a density gradient at a rate two orders of magnitude faster than the ions, the ions would be left behind and the plasma would no longer be quasi-neutral. What actually happens is that the electrons try to diffuse faster than the ions, but in so doing set up a slight charge separation. This charge separation establishes a so-called “*ambipolar*” electrostatic electric field oriented to retard the electrons and accelerate the ions, i.e., decrease the outward electron flux and increase the outward ion flux. The amount of charge separation self-regulates to produce an ambipolar electric field having just the right magnitude to equalize the outward electron and ion fluxes. This results in an effective diffusion, called the ambipolar diffusion, which is less than the electron rate, but greater than the ion rate. Equation 1.22 shows that an electric field establishes an average electron momentum $m_e \mathbf{u}_e = -e \mathbf{E} / \nu_e$, where ν_e is the rate at which the average electron loses momentum owing to collisions with ions or neutrals. Electron-electron collisions are excluded from this calculation because an “average electron” cannot lose momentum owing to collisions with all the other electrons, because the other electrons have on average the same momentum as this average electron. Since the electric field cannot impart momentum to the plasma as a whole, the momentum imparted to ions must be

equal and opposite, so $m_i \mathbf{u}_i = e \mathbf{E} / v_e$. Because diffusion in the presence of a density gradient produces an electron flux, $-D_e \nabla n_e$, the net electron flux resulting from the combination of an electric field and a diffusion across a density gradient is

$$\mathbf{\Gamma}_e = n_e \mu_e \mathbf{E} - D_e \nabla n_e, \quad (1.28)$$

where

$$\mu_e = -\frac{e}{m_e v_e} \quad (1.29)$$

is called the electron mobility. Similarly, the net ion flux is

$$\mathbf{\Gamma}_i = n_i \mu_i \mathbf{E} - D_i \nabla n_i, \quad (1.30)$$

where

$$\mu_i = \frac{e}{m_i v_i} \quad (1.31)$$

is the ion mobility. In order to maintain quasi-neutrality, the electric field automatically self-adjusts to give $\mathbf{\Gamma}_e = \mathbf{\Gamma}_i = \mathbf{\Gamma}_{ambipolar}$ and $n_i = n_e = n$; this ambipolar electric field is

$$\begin{aligned} \mathbf{E}_{ambipolar} &= \frac{(D_e - D_i)}{(\mu_e - \mu_i)} \nabla \ln n \\ &\simeq \frac{D_e}{\mu_e} \nabla \ln n \\ &= \frac{\kappa T_e}{e} \nabla \ln n. \end{aligned} \quad (1.32)$$

Substitution for $\mathbf{E}$ gives the ambipolar diffusion to be

$$\mathbf{\Gamma}_{ambipolar} = -\left(\frac{\mu_e D_i - D_e \mu_i}{\mu_e - \mu_i} \right) \nabla n, \quad (1.33)$$

so the ambipolar diffusion coefficient is

$$\begin{aligned} D_{ambipolar} &= \frac{\mu_e D_i - D_e \mu_i}{\mu_e - \mu_i} \\ &= \frac{\frac{D_i}{\mu_i} - \frac{D_e}{\mu_e}}{\frac{1}{\mu_i} - \frac{1}{\mu_e}} \\ &= \frac{D_i \frac{m_i v_i}{e} + D_e \frac{m_e v_e}{e}}{\frac{m_i v_i}{e} + \frac{m_e v_e}{e}} \\ &\simeq \frac{\kappa (T_i + T_e)}{m_i v_i}, \end{aligned} \quad (1.34)$$

where Eqs. (1.26) and (1.27) have been used as well as the relation $v_i \sim (m_e/m_i)^{1/2} v_e$. If the electrons are much hotter than the ions, then for a given ion temperature, the

ambipolar diffusion scales as T_e/m_i . Ambipolar diffusion is thus somewhat like the situation of a small child tugging on his/her parent (the energy of the small child is like the electron temperature, the parental mass is like the ion mass, and the tension in the arm that accelerates the parent and decelerates the child is like the ambipolar electric field); the resulting motion (parent and child move together faster than the parent would like and slower than the child would like) is analogous to electrons being retarded and ions being accelerated by the ambipolar electric field in such a way as to maintain quasi-neutrality.

1.12 A quantitative perspective

Relevant physical constants are

$$\begin{aligned}e &= 1.6 \times 10^{-19} \text{ C} \\m_e &= 9.1 \times 10^{-31} \text{ kg} \\m_p/m_e &= 1836 \\\epsilon_0 &= 8.85 \times 10^{-12} \text{ F m}^{-1}\end{aligned}$$

The temperature is measured in units of electron volts, so that $\kappa = 1.6 \times 10^{-19} \text{ J V}^{-1}$; i.e., $\kappa = e$. Thus, the Debye length is

$$\begin{aligned}\lambda_D &= \sqrt{\frac{\epsilon_0 \kappa T}{ne^2}} \\&= \sqrt{\frac{\epsilon_0}{e}} \sqrt{\frac{T_{eV}}{n}} \\&= 7.4 \times 10^3 \sqrt{\frac{T_{eV}}{n}} \text{ m.}\end{aligned}\tag{1.35}$$

We will assume that the typical velocity is related to the temperature by

$$\frac{1}{2}mv^2 = \frac{3}{2}\kappa T.\tag{1.36}$$

For electron-electron scattering $\mu = m_e/2$ so that the small-angle scattering cross-section is

$$\begin{aligned}\sigma^* &= \frac{1}{2\pi} \left(\frac{e^2}{\epsilon_0 mv^2/2} \right)^2 \ln(\lambda_D/b_{\pi/2}) \\&= \frac{1}{2\pi} \left(\frac{e^2}{3\epsilon_0 \kappa T} \right)^2 \ln \Lambda,\end{aligned}\tag{1.37}$$

Table 1.2 Comparison of parameters for a wide variety of plasmas.

	n	T	λ_D	$n\lambda_D^3$	$\ln \Lambda$	ν_{ee}	l_{mfp}	L
units	m^{-3}	eV	m			s^{-1}	m	m
Solar corona (loops)	10^{15}	100	10^{-3}	10^7	19	10^2	10^5	10^8
Solar wind (near Earth)	10^7	10	10	10^9	25	10^{-5}	10^{11}	10^{11}
Magnetosphere (tail lobe)	10^4	10	10^2	10^{11}	28	10^{-8}	10^{14}	10^8
Ionosphere	10^{11}	0.1	10^{-2}	10^4	14	10^2	10^3	10^5
Mag. fusion (tokamak)	10^{20}	10^4	10^{-4}	10^7	20	10^4	10^4	10
Inertial fusion (imploded)	10^{31}	10^4	10^{-10}	10^2	8	10^{14}	10^{-7}	10^{-5}
Lab plasma (dense)	10^{20}	5	10^{-6}	10^3	9	10^8	10^{-2}	10^{-1}
Lab plasma (diffuse)	10^{16}	5	10^{-4}	10^5	14	10^4	10^1	10^{-1}

where

$$\begin{aligned}\Lambda &= \frac{\lambda_D}{b_{\pi/2}} \\ &= \sqrt{\frac{\epsilon_0 \kappa T}{ne^2}} \frac{4\pi \epsilon_0 m v^2 / 2}{e^2} \\ &= 6\pi n \lambda_D^3\end{aligned}\tag{1.38}$$

is typically a very large number corresponding to there being a macroscopically large number of particles in a sphere having a radius equal to a Debye length; different authors will have slightly different numerical coefficients, depending on how they identify velocity with temperature. This difference is of no significance because of the logarithmic dependence.

The collision frequency is $\nu = \sigma^*nv$, so

$$\begin{aligned}\nu_{ee} &= \frac{n}{2\pi} \left(\frac{e^2}{3\epsilon_0 \kappa T} \right)^2 \sqrt{\frac{3\kappa T}{m_e}} \ln \Lambda \\ &= \frac{e^{5/2}}{2 \times 3^{3/2} \pi \epsilon_0^2 m_e^{1/2}} \frac{n \ln \Lambda}{T_e^{3/2}} \\ &= 4 \times 10^{-12} \frac{n \ln \Lambda}{T_e^{3/2}}.\end{aligned}\tag{1.39}$$

For most plasmas $\ln \Lambda$ lies in the range 8–25; for situations where only a simple order of magnitude estimate of the collision frequency is required it is usually sufficient to assume $\ln \Lambda \sim 10$.

Table 1.2 lists nominal parameters for several plasmas of interest and shows these plasmas have an enormous range of densities, temperatures, scale lengths,

mean free paths, and collision frequencies. The crucial issue is the ratio of the mean free path to the characteristic scale length.

Arc plasmas and magnetoplasmadynamic thrusters are in the category of dense lab plasmas; these plasmas are very collisional (the mean free path is much smaller than the characteristic scale length). The plasmas used in semiconductor processing and many research plasmas are in the diffuse lab plasma category; these plasmas are collisionless. It is possible to make both collisional and collisionless lab plasmas, and in fact if there are large temperature or density gradients it is possible to have both collisional and collisionless behavior in the same device.

1.13 Assignments

1. Rutherford scattering. This assignment involves deriving the Rutherford scattering formula using a geometrical analysis, which exploits the symmetry of the scattering trajectory.
 - (a) Show that the equation of motion in the center-of-mass frame is

$$\mu \ddot{\mathbf{r}} = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}.$$

By taking the time derivative of $\mathbf{r} \times \dot{\mathbf{r}}$ show that the angular momentum $\mathbf{L} = \mu \mathbf{r} \times \dot{\mathbf{r}}$ is a constant of the motion. The calculations will be done in the center-of-mass frame using a cylindrical coordinate system r, ϕ, z with origin at the scattering center and the z axis chosen to be parallel to $\mathbf{L}$. Show that the particle does not move in the z direction (hint: consider the dot product between $\dot{\mathbf{r}}$ and $\mathbf{L}$) so the particle is confined to the $z = 0$ plane. Let θ be the scattering angle, and let b be the impact parameter as indicated in Fig. 1.4. Also, define a Cartesian coordinate

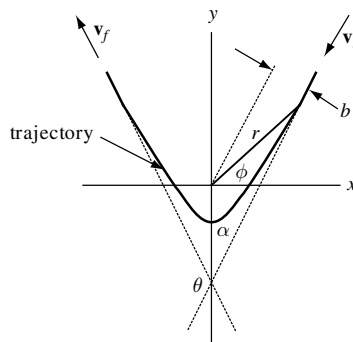


Fig. 1.4 Geometry of scattering in center-of-mass frame. Scattering center is at the origin and θ is the scattering angle. Note symmetries of velocities before and after scattering.

system x, y so that $y = r \sin \phi$, etc.; these Cartesian coordinates are also shown in Fig. 1.4.

- (b) Show that $|\mathbf{L}| = \mu b v_\infty = \mu r^2 \dot{\phi}$ so $\dot{\phi} = b v_\infty / r^2$.
- (c) Let $\mathbf{v}_i$ and $\mathbf{v}_f$ be the initial and final velocities as shown in Fig. 1.4. Since energy is conserved during scattering, the magnitudes of these two velocities must be the same, i.e., $|\mathbf{v}_i| = |\mathbf{v}_f| = v_\infty$. From the symmetry of the figure it is seen that the x component of velocity at infinity is the same before and after the collision, even though it is altered during the collision. However, the y component of the velocity reverses direction as a result of the collision. Let Δv_y be the net change in the y velocity over the entire collision. Express Δv_y in terms of v_{yi} , the y component of $\mathbf{v}_i$.
- (d) Using the y component of the equation of motion, obtain a relationship between dv_y and $d\cos \phi$. (Hint: it is useful to use conservation of angular momentum to eliminate dt in favor of $d\phi$.) Let ϕ_i and ϕ_f be the initial and final values of ϕ . By integrating dv_y , calculate Δv_y over the entire collision. How is ϕ_f related to ϕ_i and to α (refer to figure)?
- (e) How is v_{yi} related to ϕ_i and v_∞ ? How is θ related to α ? Use the expressions for Δv_y obtained in parts (c) and (d) above to obtain the Rutherford scattering formula

$$\tan\left(\frac{\theta}{2}\right) = \frac{q_1 q_2}{4\pi\epsilon_0 \mu b v_\infty^2}.$$

What is the scattering angle for grazing (small-angle collisions) and how does this small-angle scattering relate to the initial center-of-mass kinetic energy and to the potential energy at distance b ? For grazing collisions how does b relate to the distance of closest approach? What impact parameter gives 90° scattering?

2. One-dimensional scattering relations. The separation of collision types according to m_e/m_i can also be understood by considering how the combination of conservation of momentum and of energy constrain certain properties of collisions. Suppose a particle with mass m_1 and incident velocity v_1 makes a head-on collision with a stationary target particle having mass m_2 . The conservation equations for momentum and energy can be written as

$$\begin{aligned} m_1 v_1 &= m_1 v'_1 + m_2 v'_2 \\ \frac{1}{2} m_1 v_1^2 &= \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2, \end{aligned}$$

where prime refers to the value after the collision. By eliminating v'_1 between these two equations obtain v'_2 as a function of v_1 . Use this to construct an expression showing the ratio $m_2 v_2'^2 / m_1 v_1^2$, i.e., the fraction of the incident particle energy transferred to the target particle as a result of the collision. How does this fraction depend on m_1/m_2 when m_1/m_2 is equal to unity, very large, or very small? If m_1/m_2 is very large or very small how many collisions are required to transfer approximately all of the incident particle energy to target particles?

3. Some basic facts you should know. Memorize the value of ϵ_0 (or else arrange for the value to be close at hand). What is the value of Boltzmann's constant when temperatures are measured in electron volts? What is the density of the air you are breathing, measured in particles per cubic meter? What is the density of particles in solid copper, measured in particles per cubic meter? What is room temperature, expressed in electron volts? What is the ionization potential (in eV) of a hydrogen atom? What is the mass of an electron and of an ion (in kilograms)? What is the strength of the Earth's magnetic field at your location, expressed in Tesla? What is the strength of the magnetic field produced by a straight wire carrying 1 ampere as measured by an observer located 1 meter from the wire and what is the direction of the magnetic field? What is the relationship between Tesla and Gauss, between particles per cubic centimeter and particles per cubic meter? What is magnetic flux? If a circular loop of wire with a break in it links a magnetic flux of 29.83 Wb, which increases at a constant rate to a flux of 30.83 Wb, in one second, what voltage appears across the break?
4. Solve Eq. (1.6) the "easy" way using Gauss' law to show that the solution of

$$\nabla^2 \phi = -\frac{1}{\epsilon_0} \delta(\mathbf{r})$$

is

$$\phi = \frac{1}{4\pi\epsilon_0 r}.$$

Show that this implies

$$\nabla^2 \frac{1}{4\pi r} = -\delta(\mathbf{r}) \quad (1.40)$$

is a representation for the delta function. Then, use spherical polar coordinates and symmetry to show the Laplacian reduces to

$$\nabla^2 \phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right).$$

Explicitly calculate $\nabla^2(1/r)$ and then reconcile your result with Eq. (1.40). Using these results guess that the solution to Eq. (1.6) has the form

$$\phi = \frac{g(r)}{4\pi\epsilon_0 r}.$$

Substitute this guess into Eq. (1.6) to obtain a differential equation for g that is trivial to solve.

5. Solve Eq. (1.6) for $\phi(\mathbf{r})$ using a more general method that illustrates several important mathematical techniques and formalisms. Begin by defining the 3-D Fourier transform

$$\tilde{\phi}(\mathbf{k}) = \int d\mathbf{r} \phi(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}}, \quad (1.41)$$

in which case the inverse transform is

$$\phi(\mathbf{r}) = \frac{1}{(2\pi)^3} \int d\mathbf{k} \tilde{\phi}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} \quad (1.42)$$

and note that the Dirac delta function can be expressed as

$$\delta(\mathbf{r}) = \frac{1}{(2\pi)^3} \int d\mathbf{k} e^{i\mathbf{k} \cdot \mathbf{r}}. \quad (1.43)$$

Now multiply Eq. (1.6) by $\exp(-i\mathbf{k} \cdot \mathbf{r})$ and then integrate over all $\mathbf{r}$, i.e., operate with $\int d\mathbf{r}$. The term involving ∇^2 is integrated by parts, which effectively replaces the ∇ operator with $i\mathbf{k}$.

Show that the Fourier transform of the potential is

$$\tilde{\phi}(\mathbf{k}) = \frac{q_T}{\epsilon_0(k^2 + \lambda_D^{-2})} \quad (1.44)$$

and use this in Eq. (1.42).

Because of spherical symmetry use spherical polar coordinates for the k -space integral. The only fixed direction is the $\mathbf{r}$ direction so choose the polar axis of the $\mathbf{k}$ coordinate system to be parallel to $\mathbf{r}$. Thus $\mathbf{k} \cdot \mathbf{r} = kr\alpha$, where $\alpha = \cos \theta$ and θ is the polar angle. Also, $d\mathbf{k} = -d\phi k^2 d\alpha dk$, where ϕ is the azimuthal angle. What are the limits of the respective ϕ , α , and k integrals? In answering this, you should first obtain an integral of the form

$$\phi(\mathbf{r}) \sim \int_{\phi=?}^? d\phi \int_{\alpha=?}^? d\alpha \int_{k=?}^? k^2 dk \times (?), \quad (1.45)$$

where the limits and the integrand with appropriate coefficients are specified (i.e., replace all the question marks and $\sim$ by the correct quantities). Upon evaluation of the ϕ and α integrals Eq. (1.45) becomes an even function of k so that the range of integration can be extended to $-\infty$ providing the overall integral is multiplied by $1/2$. Realizing that $\sin kr = \text{Im} [e^{ikr}]$, derive an expression of the general form

$$\phi(\mathbf{r}) \sim \text{Im} \int_{-\infty}^{\infty} k dk \frac{e^{ikr}}{f(k^2)} \quad (1.46)$$

but specify the coefficient and exact form of $f(k^2)$. Explain why the integration contour (which is along the real k axis) can be completed in the upper half complex k -plane. Complete the contour in the upper half-plane and show that the integrand has a single pole in the upper half-plane at $k = ?$ Use the method of residues to obtain $\phi(\mathbf{r})$.

6. Make sure you know how to evaluate quickly $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$ and $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$. A useful mnemonic, which works for both forms, is: “both forms = middle times (other two dotted together) – outer times (other two dotted together),” where outer refers to the outer vector of the parentheses (furthest from the center of the triad) and middle refers to the middle vector in the triad of vectors; see Appendix A.
7. Particle trajectory integration scheme (Birdsall and Langdon 1985). In this assignment you will develop a simple, but powerful, “leap-frog” numerical integration scheme. This is a type of “implicit” numerical integration scheme. This numerical scheme can

later be used to evaluate particle orbits in time-dependent fields having complex topology. These calculations can be considered as numerical experiments used in conjunction with the analytic theory we will develop. This combined analytical/numerical approach provides a deeper insight into charged particle dynamics than analysis alone.

Brief note on Implicit vs. Explicit numerical integration schemes

Suppose it is desired to use numerical methods to integrate the equation

$$\frac{dy}{dt} = f(y(t), t).$$

Unfortunately, since $y(t)$ is the sought-after quantity, we do not know what to use in the right-hand side for $y(t)$. A naive choice would be to use the previous value of y in the right-hand side to get a scheme of the form

$$\frac{y_{\text{new}} - y_{\text{old}}}{\Delta t} = f(y_{\text{old}}, t),$$

which may be solved to give

$$y_{\text{new}} = y_{\text{old}} + \Delta t f(y_{\text{old}}, t).$$

Simple and appealing as this is, it does not work since it is numerically unstable. However, if we use the following scheme we will get a stable result:

$$\frac{y_{\text{new}} - y_{\text{old}}}{\Delta t} = f((y_{\text{new}} + y_{\text{old}})/2, t). \quad (1.47)$$

In other words, we have used the average of the new and the old values of y in the right-hand side. This makes sense because the right-hand side is a function evaluated at time t , whereas $y_{\text{new}} = y(t + \Delta t/2)$ and $y_{\text{old}} = y(t - \Delta t/2)$. If these last quantities are Taylor expanded, it is seen that to lowest order $y(t) = [y(t + \Delta t/2) + y(t - \Delta t/2)]/2$. Since y_{new} occurs on both sides of the equation we will have to solve some sort of equation, or invert some sort of matrix, to get y_{new} . Start with

$$m \frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Define the angular cyclotron frequency vector $\boldsymbol{\Omega} = q\mathbf{B}/m$ and the normalized electric field $\boldsymbol{\Sigma} = q\mathbf{E}/m$ so that the above equation becomes

$$\frac{d\mathbf{v}}{dt} = \boldsymbol{\Sigma} + \mathbf{v} \times \boldsymbol{\Omega}. \quad (1.48)$$

Using the implicit scheme of Eq. (1.47), show that Eq. (1.48) becomes

$$\mathbf{v}_{\text{new}} + \mathbf{A} \times \mathbf{v}_{\text{new}} = \mathbf{C},$$

where $\mathbf{A} = \boldsymbol{\Omega}\Delta t/2$ and $\mathbf{C} = \mathbf{v}_{\text{old}} + \Delta t (\boldsymbol{\Sigma} + \mathbf{v}_{\text{old}} \times \boldsymbol{\Omega}/2)$. By first dotting the above equation with $\mathbf{A}$ and then crossing it with $\mathbf{A}$ show that the new value of velocity is given by

$$\mathbf{v}_{\text{new}} = \frac{\mathbf{C} + \mathbf{A}\mathbf{A} \cdot \mathbf{C} - \mathbf{A} \times \mathbf{C}}{1 + A^2}.$$

The new position is simply given by

$$\mathbf{x}_{\text{new}} = \mathbf{x}_{\text{old}} + \mathbf{v}_{\text{new}} dt.$$

The above two equations can be used to solve charged particle motion in complicated, 3-D, time-dependent fields. Use this particle integrator to calculate and plot the trajectory of an electron moving in crossed electric and magnetic fields where the non-vanishing components are $E_x = 1 \text{ V m}^{-1}$ and $B_z = 1 \text{ T}$. Try varying the field strengths and polarities, and also try ions instead of electrons.

8. Use the leap-frog numerical integration scheme to demonstrate the Rutherford scattering problem:

(i) Define a characteristic length for this problem to be the impact parameter for a 90° degree scattering angle, $b_{\pi/2}$. A reasonable choice for the characteristic velocity is v_∞ . What is the characteristic time?

(ii) Define a Cartesian coordinate system such that the z axis is parallel to the incident relative velocity vector $\mathbf{v}_\infty$ and goes through the scattering center. Let the impact parameter be in the y direction so that the incident particle is traveling in the $y-z$ plane. Establish a plotting frame spanning the region $-50 \leq z/b_{\pi/2} \leq 50$ and $-50 \leq y/b_{\pi/2} \leq 50$.

(iii) Set the magnetic field to be zero, and let the electric field be

$$\mathbf{E} = -\nabla\phi,$$

where $\phi = ?$ so $E_x = ?$, etc.

(iv) By using $r^2 = x^2 + y^2 + z^2$ calculate the electric field at each particle position, and so determine the particle trajectory.

(v) Demonstrate that the scattering is indeed at 90° when $b = b_{\pi/2}$. What happens when b is much larger or much smaller than $b_{\pi/2}$? What happens when q_1, q_2 have the same or opposite signs?

(vi) Have your code draw the relevant theoretical scattering angle θ_s and show that the numerical result is in agreement with the theoretical prediction.

9. Collision relations. Show that $\sigma n_t l_{mfp} = 1$, where σ is the cross-section for a collision, n_t is the density of target particles, and l_{mfp} is the mean free path. Show also that the collision frequency is given by $\nu = \sigma n_t v$, where v is the velocity of the incident particle. Calculate the electron-electron collision frequency for the following plasmas: fusion ($n \sim 10^{20} \text{ m}^{-3}$, $T \sim 10 \text{ keV}$), partially ionized discharge plasma ($n \sim 10^{16} \text{ m}^{-3}$, $T \sim 10 \text{ eV}$). At what temperature does the conductivity of plasma equal that of copper, and of steel? Assume $Z = 1$.

10. Cyclotron motion. Suppose that a particle is immersed in a uniform magnetic field $\mathbf{B} = B\hat{z}$ and there is no electric field. Suppose that at $t = 0$ the particle's initial position is at $\mathbf{x} = 0$ and its initial velocity is $\mathbf{v} = v_0\hat{x}$. Using the Lorentz equation, calculate the particle position and velocity as a function of time (be sure to take initial conditions into account). What is the direction of rotation for ions and for electrons (right handed or left handed with respect to the magnetic field)? If you had to make up a mnemonic for the sense of ion rotation, would it be Lions or Rions? Now, repeat the analysis but this time with an electric field $\mathbf{E} = \hat{x}E_0\cos(\omega t)$. What happens in the limit where $\omega \rightarrow \Omega$, where $\Omega = qB/m$ is the cyclotron frequency? Assume that the particle is a proton and that $B = 1\text{ T}$, $v_0 = 10^5\text{ m s}^{-1}$, and compare your results with direct numerical solution of the Lorentz equation. Use $E_0 = 10^4\text{ V m}^{-1}$ for the electric field.
11. Space-charge-limited current. When a metal or metal oxide is heated to high temperatures it emits electrons from its surface. This process, called *thermionic emission*, is the basis of vacuum tube technology and is also essential when high currents are drawn from electrodes in a plasma. The electron-emitting electrode is called a *cathode*, while the electrode to which the electrons flow is called an *anode*. An idealized configuration is shown in Fig. 1.5. This configuration can operate in two regimes: (i) the *temperature-limited* regime, where the current is determined by the thermionic emission capability of the cathode, and (ii) the *space-charge-limited* regime, where the current is determined by a buildup of electron density in the region between cathode and anode (inter-electrode region). Let us now discuss this space-charge-limited regime: if the current is small, then the number of electrons required to carry the current is small and so the inter-electrode region is nearly a vacuum, in which case the electric field in this region will be nearly uniform and be given by $E = V/d$, where V is the anode–cathode potential difference and d is the anode–cathode separation. This electric field will accelerate the electrons from anode to cathode. However, if the current is large, there will be a significant electron density in the inter-electrode

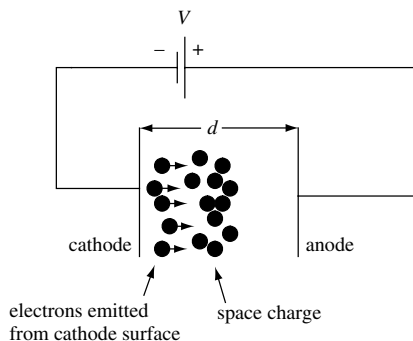


Fig. 1.5 Electron cloud accelerated from cathode to anode encounters space charge of previously emitted electrons.

region. This “space charge” will create a localized depression in the potential (since electrons have negative charge).

The result is that the electric field will be reduced in the region near the cathode. If the space charge is sufficiently large, the electric field at the cathode vanishes. In this situation, attempting to increase the current by increasing the number of electrons ejected by the cathode will not succeed because an increase in current (which will give an increase in space charge) will produce a repulsive electric field that prevents additional electrons from leaving the cathode. Let us now calculate the space-charge-limited current and relate it to our discussion on Debye shielding. The current density in this system is

$$J = -n(x)ev(x) = \text{a negative constant.}$$

Since potential is undefined with respect to a constant, let us choose this constant so that the cathode potential is zero, in which case the anode potential is V_0 . Assuming electrons leave the cathode with zero velocity, show that the dependence of electron velocity on position is given by

$$v(x) = \sqrt{\frac{2eV(x)}{m_e}}.$$

Show that the above two equations, plus Poisson’s equation, can be combined to give the following differential equation for the potential:

$$\frac{d^2V}{dx^2} - \lambda V^{-1/2} = 0,$$

where $\lambda = \varepsilon_0^{-1} |J| \sqrt{m_e/2e}$. By multiplying this equation with the integrating factor dV/dx and using the space-charge-limited boundary condition $E = 0$ at $x = 0$, solve for $V(x)$. By rearranging the expression for $V(x)$ show that the space-charge-limited current is

$$J = \frac{4}{9} \varepsilon_0 \sqrt{\frac{2e}{m_e}} \frac{V^{3/2}}{d^2}.$$

This is called the Child–Langmuir space charge limited current. For reference the temperature-limited current is given by the Richardson–Dushman law,

$$J = AT^2 e^{-\phi_0/\kappa T},$$

where the coefficient A and the *work function* ϕ_0 are properties of the cathode material, while T is the cathode temperature. Thus, the actual cathode current will be whichever is the smaller of the above two expressions. Show there is a close relationship between the physics underlying the Child–Langmuir law and Debye shielding (hint: characterize the electron velocity as being a thermal velocity and its energy as being a thermal energy and then show that the inter-electrode spacing corresponds to λ_D). Suppose that a cathode was operating in the space-charge-limited regime and that some positively charged ions were placed in the inter-electrode region. What would happen to the space charge – would it be possible to draw

more or less current from the cathode? Suppose the entire inter-electrode region were filled with plasma with electron temperature T_e . What would be the appropriate value of d and how much current could be drawn from the cathode (assuming it were sufficiently hot)? Does this give you any ideas on why high current switch tubes (called ignitrons) use plasma to conduct the current?